PWM Motor Control Electrical Validation Report

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1. Objective

The objective of this project is to design and electrically validate a pulse width modulation (PWM) motor control circuit using an Arduino Uno and a MOSFET (IRLZ44N). The system will be characterized using oscilloscope and multimeter measurements to evaluate PWM frequency accuracy, duty cycle accuracy, switching behavior, and inductive transient response under motor load conditions. Measured results will be compared to theoretical expectations to assess signal integrity and protection effectiveness.

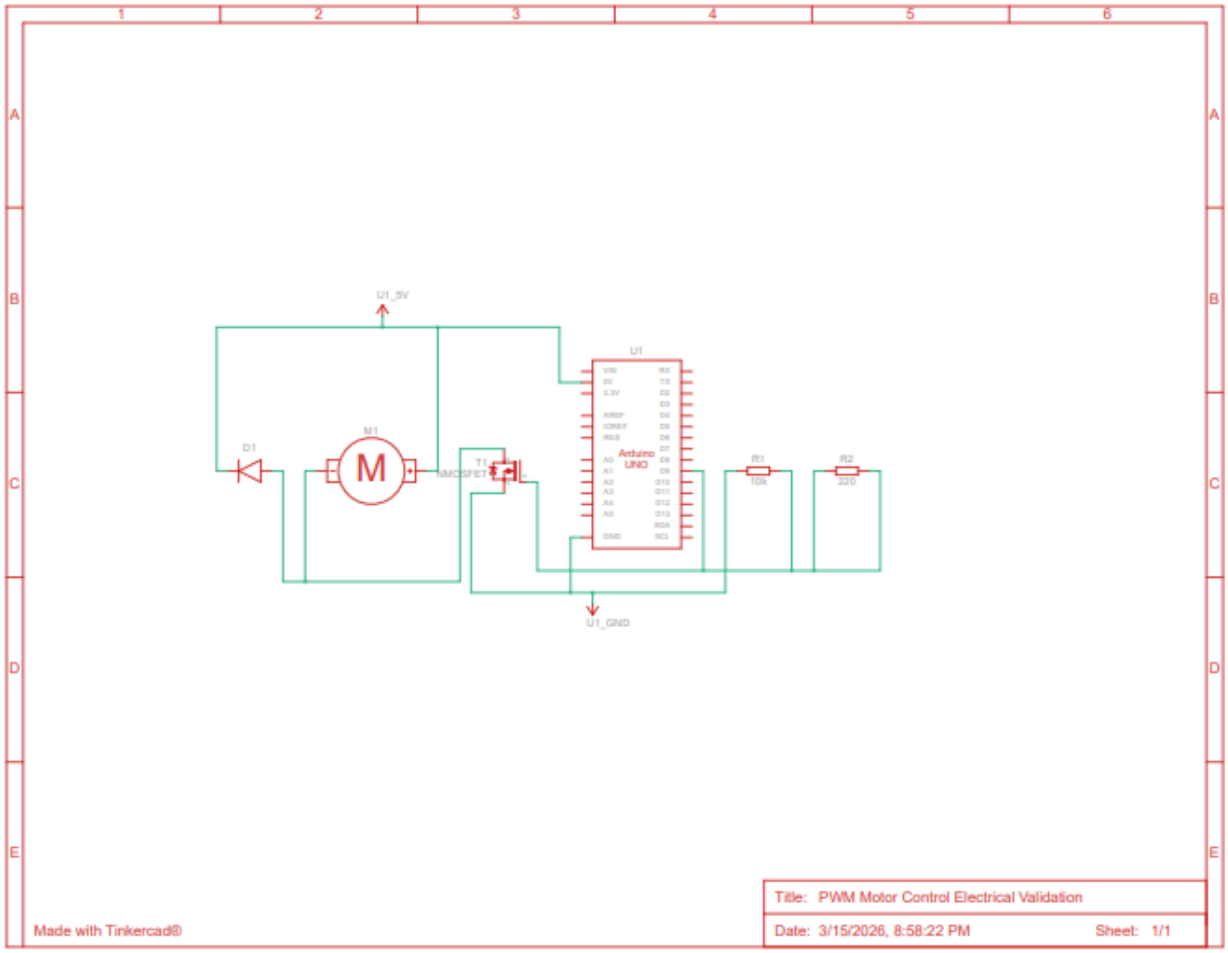
2. System Overview

The system consists of an Arduino Uno generating a PWM control signal to drive a DC motor through an IRLZ44N logic level N-channel MOSFET. The MOSFET operates as a low side switch, allowing the Arduino to control motor speed by varying the PWM duty cycle. A flyback diode is placed across the motor terminals to suppress inductive voltage transients generated during switching. PWM control regulates motor speed by rapidly switching the supply voltage on and off at a fixed frequency while varying the percentage of time the signal remains high (duty cycle). Due to mechanical inertia and inductive characteristics, the motor responds to the average voltage of the PWM signal rather than individual pulses. The system is evaluated in multiple stages, beginning with PWM signal validation without load, followed by integration of the MOSFET switching stage, and finally full motor load testing with transient analysis.

3. Test Equipment

- Arduino Uno
- IRLZ44N MOSFET
- DC Motor (3-6V) with fan blade
- 1N4007 Rectifier Diode
- 220 Ω, 10k Ω Resistors
- SK201B+ Digital Oscilloscope/Multimeter

4. Hardware Schematic



5. Test Setup

Stage 1

Initial validation was performed on the Arduino PWM output prior to integrating the MOSFET switching stage or motor load. The PWM signal was generated from Arduino digital pin 9 using the analogWrite() function with a value of 128, corresponding to a 50% duty cycle. The oscilloscope probe was connected directly to the PWM output pin with the ground reference connected to the Arduino ground. This configuration allowed direct observation of the raw PWM waveform generated by the microcontroller. Oscilloscope settings used during measurement included:

- Time base: 1 ms/div
- Voltage Scale: 3 V/div
- Trigger Mode: Rising edge trigger
- Trigger Level: 2.5V

Stage 2

Following verification of the PWM signal generator, the MOSFET switching stage and DC motor were integrated into the circuit. The PWM output from the Arduino Uno digital pin 9 was connected to the gate of the IRLZ44N MOSFET through a 220Ω resistor. The MOSFET was configured as a low side switch, with the source connected to ground and the drain connected to one terminal of the DC motor. The opposite motor terminal was connected to the Arduino 5 V supply rail. A 1N4007 Rectifier Diode was placed across the motor terminals with the cathode connected to the 5 V rail and the anode connected to the MOSFET drain. This diode serves as a flyback protection device to suppress inductive voltage spikes generated when the motor current is rapidly interrupted during PWM switching. During Stage 2 testing, the PWM duty cycle was varied to observe the effect on motor operation and to validate the relationship between duty cycle and effective average voltage applied to the motor.

Stage 3

The final stage of testing focused on evaluating MOSFET switching behavior and observing inductive transient effects generated by the DC motor load. The motor and MOSFET switching circuit remained configured as described in the previous stage. The oscilloscope probe was moved from the MOSFET gate to the MOSFET drain node, which corresponds to the switching point between the motor and the MOSFET. This location allows observation of the voltage waveform produced as the MOSFET rapidly switches the motor current on and off. Because the motor behaves as an inductive load, rapid switching can generate voltage transients when the current path is interrupted. To mitigate this effect, a 1N4007 Rectifier Diode was placed across the motor terminals to provide a safe discharge path for the stored inductive energy.

6. Test Procedure

Stage 1

An Arduino sketch was uploaded to generate a PWM signal using the analogWrite() function on digital pin 9 with a value of 128 (50% duty cycle).
The oscilloscope alligator probe tip was connected to the PWM output pin and the ground clip was connected to Arduino ground.
Oscilloscope scaling and trigger settings were adjusted to clearly visualize the PWM waveform.
Peak to peak voltage, average voltage, and RMS voltage were recorded directly from the oscilloscope display.
PWM frequency was estimated by counting waveform cycles across the oscilloscope time base.

Stage 2

1. A new Arduino test program was uploaded to vary the PWM duty cycle across several discrete values using the analogWrite() function on digital pin 9. The duty cycle was sequentially set to approximately 25%, 50%, 75%, and 100% by writing values of 64, 128, 192, and 255 respectively.
2. The oscilloscope probe was connected to the MOSFET gate to observe the PWM waveform.
3. Peak to peak voltage, average voltage, RMS voltage, and frequency were recorded from the oscilloscope display.
4. The motor response was visually observed and qualitatively categorized based on relative rotational speed.

Stage 3

1. Executed an Arduino program to drive the motor using PWM at a fixed duty cycle. The PWM signal was generated on digital pin 9 using the analogWrite() function with a value of 128 (50% duty cycle).
2. While the PWM signal runs, the oscilloscope probe is connected to the MOSFET drain node while the ground reference is connected to circuit ground.
3. Waveform stability was recorded from the oscilloscope display.

7. Results

Stage 1

Parameter	Expected Value	Measured Value
Peak to Peak Voltage	5 V	4.92 V
Average Voltage	2.5 V	2.39 V
RMS Voltage	2.70 V	2.70 V
PWM Frequency	490 Hz	500 Hz*

* Frequency was estimated by observing approximately five waveform cycles across a 10 ms oscilloscope window.

Stage 2

Duty Cycle	Peak to Peak Voltage	Average Voltage	RMS Voltage	Frequency
25%	4.15 Vp-p	1.08 V	1.49 V	490 Hz
50%	4.56 Vp-p	2.25 V	2.56 V	490 Hz
75%	4.12 Vp-p	3.23 V	3.43 V	490 Hz
100%	4.78 Vp-p	4.18 V	4.38 V	490 Hz

Stage 3

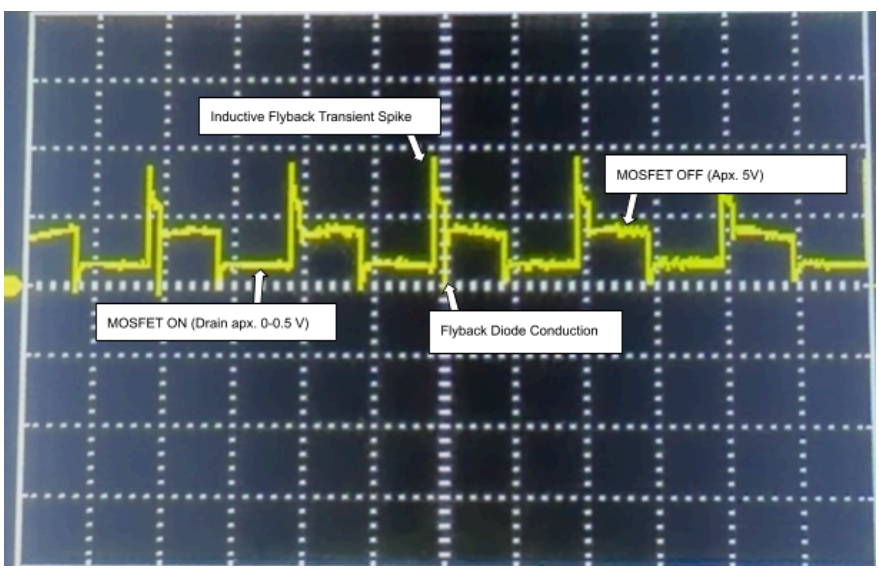


Figure 1: MOSFET Drain Switching Waveform

8. Analysis

Stage 1

The measured PWM waveform closely matched theoretical expectations for a 50% duty cycle signal generated by the Arduino Uno. The peak to peak voltage measured 4.92 V, which is consistent with the Arduino's nominal 5 V logic level output.
The measured average voltage of 2.39 V is slightly lower than the theoretical value of 2.5 V for a 50% duty cycle waveform, which may be attributed to measurement tolerance, oscilloscope resolution, and minor signal loading effects.
The estimated PWM frequency of approximately 500 Hz closely matches the expected Arduino PWM frequency of approximately 490 Hz on pin 9. The waveform displayed stable switching behavior with clean rising and falling edges. These results confirm that the Arduino PWM signal generator is functioning correctly and is suitable for driving the MOSFET switching stage in subsequent testing.

Stage 2

The results demonstrate the expected relationship between PWM duty cycle and motor speed. As the duty cycle increased, the proportion of time that the PWM signal remained in the high state increased, resulting in a higher effective average voltage applied across the motor terminals. Because the DC motor functions with both inductive and mechanical inertia, it responds primarily to the average voltage of the PWM waveform rather than the instantaneous switching transitions of the PWM signal. Increasing the duty cycle resulted in progressively higher motor speeds. The PWM frequency remained approximately constant throughout testing, confirming that the microcontroller maintained a fixed switching frequency while modulating the duty cycle. The peak to peak voltage also remained consistent near the 5 V logic level output of the Arduino, indicating stable MOSFET gate drive conditions. These results confirm proper operation of the MOSFET switching stage and demonstrate successful control of motor speed using PWM duty cycles.

Stage 3

The waveform observed from the oscilloscope from the MOSFET drain demonstrated the expected switching behavior of a PWM driven inductive load, as shown in Figure 1. When the MOSFET turned on, the drain voltage dropped close to ground as current flowed through the motor. When the MOSFET switched off, the voltage rose back toward the motor supply level. Small voltage spikes were visible at the switching transitions due to the DC motor, which is a characteristic of an inductive load. These transients occur as the magnetic field within the motor windings collapses when current flow is interrupted. The flyback diode effectively limited the magnitude of these spikes by providing a path for the stored energy to dissipate safely. This can be seen by the subsequent dip before stabilizing to an expected value. The switching transitions remained stable and repeatable throughout testing, indicating reliable MOSFET operation. These results confirm that the circuit successfully controls the motor using PWM while properly managing inductive transients through the flyback protection diode.

9. Measurement Limitations

The SK201B+ Digital Oscilloscope Multimeter used in this experiment has limited bandwidth and sampling resolution compared to laboratory-grade oscilloscopes, which may introduce small measurement inaccuracies in voltage and frequency readings. Additional variation may result from breadboard contact resistance, probe grounding position, and electrical noise generated by the motor during switching. Despite these limitations, the measured results remained close to theoretical expectations and clearly demonstrated the intended PWM switching behavior of the circuit.

10. Conclusion

This project successfully demonstrated the design and electrical validation of a PWM controlled DC motor driver using an Arduino Uno and an IRLZ44N MOSFET. Initial testing confirmed that the Arduino generated a stable PWM signal with the expected frequency and voltage measurements. Integration of the MOSFET into the circuit allowed effective control of motor speed through changes in PWM duty cycle. Final measurements at the MOSFET drain revealed the expected switching waveform and transient behavior produced by the inductive motor load. The flyback diode effectively suppressed voltage spikes generated during switching transitions, protecting the MOSFET and maintaining stable circuit operation. Overall, the measured results closely matched theoretical expectations and validated the effectiveness of PWM motor control using a MOSFET switching stage.

Several improvements could be made for this project. Using an external power supply would allow testing with higher voltage motors and greater load currents, enabling evaluation of the MOSFET's performance under higher power conditions. Additional measurements could also be performed to analyze MOSFET switching losses and thermal behavior during continuous operation. A higher bandwidth oscilloscope would allow more detailed observation of transient switching events. Finally,using a feedback sensor or encoder could allow the system to automatically regulate motor speed rather than relying solely on open loop PWM control.